A3-PS003-TC207

Federated Traffic-Flow Prediction with Communication-Cost Evaluation.

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Abstract

Problem Description

Clear outline of the problem being addressed. Use bullet points for key issues.

Centralized traffic forecasting requires edge cameras and roadside sensors to continuously stream raw data to a central cloud, consuming high cellular bandwidth and concentrating private travel logs. Federated Learning addresses this by training local models on edge devices and sharing only mathematical weight updates with a central server. In this project, you will build a federated simulator using a public traffic dataset partitioned across virtual

intersections. You will evaluate how well a compact forecasting model using Federated Averaging (FedAvg) handles non-identically distributed (non-IID) traffic patterns, reporting prediction accuracy (MAE and RMSE) against network byte transmission costs per training round.

Background

Background

It should capture the background necessary to understand the problem and the proposed optimization

Traffic-flow prediction is an important part of smart city development. It helps monitor traffic conditions, predict congestion, and improve traffic management. Traditional centralized learning collects traffic data from different road intersections and sends it to a central server

for model training. However, this approach requires significant network bandwidth and transfers raw traffic data.

Federated Learning (FL) is a distributed machine learning technique that allows multiple clients to train a model locally without sharing their raw data. Instead, clients send model updates to a central server, which combines them using Federated Averaging (FedAvg).

Literature & RFC Review

Do a detailed study of exisiting proposals, studies and standards. But only mention the most relevant ones

Literature Review: Federated Learning (FL) trains a shared model without sharing raw data. FedAvg reduces communication by combining local model updates. Traffic prediction is evaluated using MAE, RMSE, and communication cost (bytes transferred).

- RFC Review:

RFC 9293 – TCP

RFC 768 – UDP

RFC 791 – IP

- References: McMahan et al. (2017), Kairouz et al. (2021), and RFC 9293, RFC 768, RFC 791.

Proposed Solution

Detailed description of the overall solution.

Use Federated Learning to predict traffic flow across multiple locations without sharing raw traffic data. Evaluate prediction accuracy and communication cost by measuring data transferred between clients and the central server.

Optimization Proposed

List of optimizations and enhancements. Use a table to compare before and after scenarios.

The proposed system optimizes federated learning to reduce communication costs while maintaining high traffic prediction accuracy. Model updates are compressed before transmission to reduce the amount of data exchanged between clients and the central server. The number of communication rounds is reduced by sending updates at suitable

intervals. Efficient client selection is used to choose clients with useful traffic data for model training. These techniques help minimize bandwidth usage, reduce communication overhead, and improve the overall efficiency of federated traffic-flow prediction.

Solution Architecture & Design

Visual: Flowchart or diagram showing the optimization process.

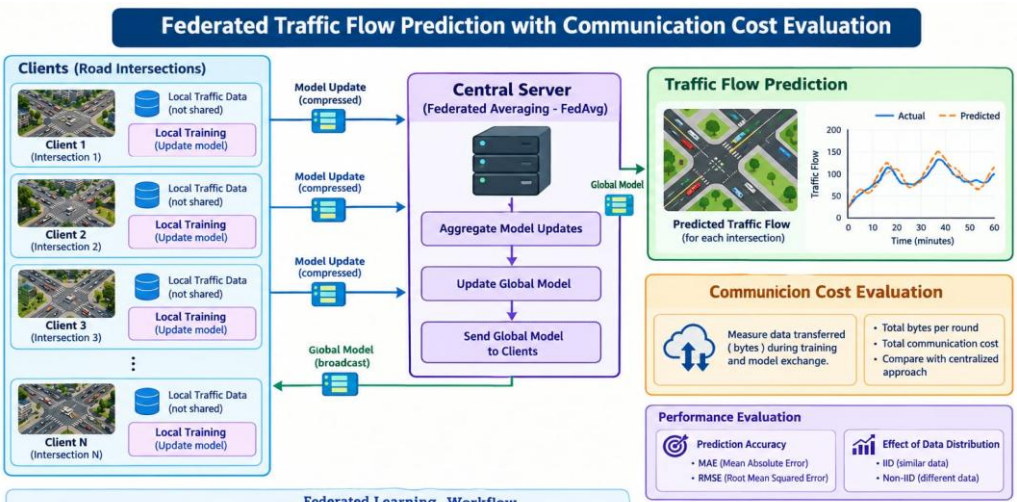
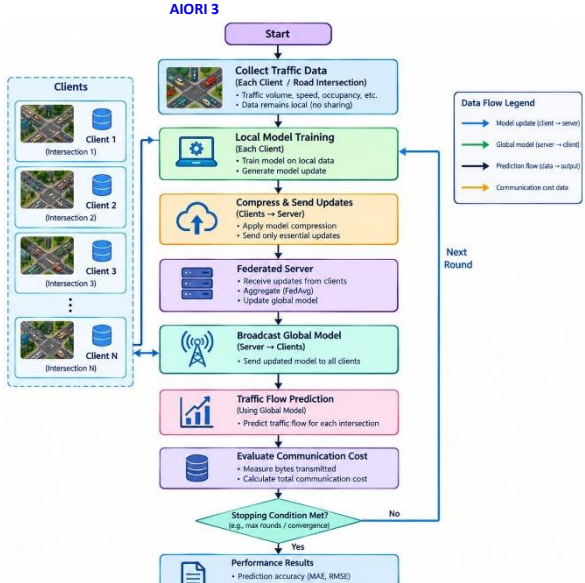

Federated Traffic Flow Prediction with Communication Cost Evaluation
Clients (Road Intersections)
Client 1
(Intersection 1)
Local Traffic Data
(not shared)
Local Training
(Update model)
Client 2
(Intersection 2)
Local Traffic Data
(not shared)
Local Training
(Update model)
Client 3
(Intersection 3)
Local Traffic Data
(not shared)
Local Training
(Update model)
Client N
(Intersection N)
Local Traffic Data
(not shared)
Local Training
(Update model)
Model Update
(compressed)
Model Update
(compressed)
Model Update
(compressed)
Global Model
(broadcast)
Central Server
(Federated Averaging - FedAvg)
Aggregate Model Updates
Update Global Model
Send Global Model
to Clients
Global Model
Traffic Flow Prediction
Predicted Traffic Flow
(for each intersection)
Actual
Predicted
Traffic Flow
Time (minutes)
Communicion Cost Evaluation
Measure data transferred
(bytes) during training
and model exchange.
• Total bytes per round
• Total communication cost
• Compare with centralized
approach
Performance Evaluation
Prediction Accuracy
• MAE (Mean Absolute Error)
• RMSE (Root Mean Squared Error)
Effect of Data Distribution
• IID (similar data)
• Non-IID (different data)
Federated Learning Workflow
1. Local Training
(at each client)
2. Send Compressed
Updates
3. Aggregate
(FedAvg)
4. Distribute Global Model
(to clients)
5. Evaluate
(Accuracy + Communication Cost)

Figure: System Architecture

Data Flow Diagram

Overview of the architecture. Include the flow chart diagram created earlier. Visual: Diagrams or architecture visuals.


AIORI 3
Start
Collect Traffic Data
(Each Client / Road Intersection)
• Traffic volume, speed, occupancy, etc.
• Data remains local (no sharing)
Clients
Client 1
(Intersection 1)
Client 2
(Intersection 2)
Client 3
(Intersection 3)
Client N
(Intersection N)
Local Model Training
(Each Client)
• Train model on local data
• Generate model update
Compress & Send Updates
(Clients → Server)
• Apply model compression
• Send only essential updates
Federated Server
• Receive updates from clients
• Aggregate (FedAvg)
• Update global model
Broadcast Global Model
(Server → Clients)
• Send updated model to all clients
Traffic Flow Prediction
(Using Global Model)
• Predict traffic flow for each intersection
Evaluate Communication Cost
• Measure bytes transmitted
• Calculate total communication cost
Stopping Condition Met?
(e.g., max rounds / convergence)
No
Yes
Next Round
Performance Results
• Prediction accuracy (MAE, RMSE)
Data Flow Legend
Model update (client → server)
Global model (server → client)
Prediction flow (data → output)
Communication cost data

Deployment Timeline

Clearly define the deployment milestones and methodology.

The project will be completed in eight weeks. The first two weeks will focus on literature review and data collection. Weeks three and four will involve developing traffic prediction

models and implementing federated learning. In week five, communication cost optimization will be performed. Week six will focus on evaluating prediction accuracy and communication cost. Week seven will involve testing and deployment, and week eight will be dedicated to documentation and final presentation.

Bibliography

Bibliography I

Bibliography

Project: Federated Traffic Flow Prediction with Communication Cost Evaluation

The bibliography lists the research papers and technical references used to develop the project on federated learning, traffic prediction, and communication cost reduction.

1\. McMahan, B. et al. (2017). Communication-Efficient Learning of Deep Networks from Decentralized Data. AISTATS.

2\. Kairouz, P. et al. (2021). Advances and Open Problems in Federated Learning.

3\. Communication-Efficient Personalized Federated Learning for Traffic Flow Prediction. IEEE International Conference on Computer and Communications, 2024.

4\. Internet Engineering Task Force (IETF). RFC 9293 – Transmission Control Protocol (TCP).

5\. Internet Engineering Task Force (IETF). RFC 768 – User Datagram Protocol (UDP).

6\. Internet Engineering Task Force (IETF). RFC 791 – Internet Protocol (IP).

7\. Research materials on traffic-flow prediction, federated learning, model compression, and communication-cost evaluation.

Thank You!